3.5 Graphing Linear Equations in Standard Form

Name _____ Period ____ Date _____

Algebra I 3.5 (30-minute plan) · Student sheet

I can…

- graph horizontal and vertical lines.
- find the intercepts of Ax + By = C and use them to graph.
- interpret the intercepts in real-life situations.

Vocabulary · fill in the missing word

Standard form — Ax + By = _____, where A and B are not both 0

Horizontal line — y = _____

Vertical line — x = a (not a _____)

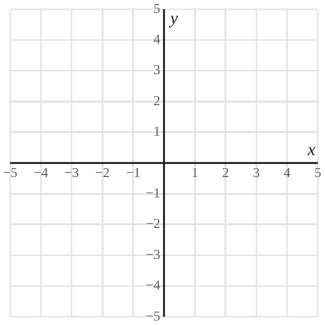
Intercepts — _____: let y = 0; y-intercept: let x = 0

Part 1 · Notes and examples (fill in each note, then work the example)

Horizontal and vertical lines: y = b is a _____ line; x = a is a _____ line.

Example 1

Graph y = −3 and x = 2.

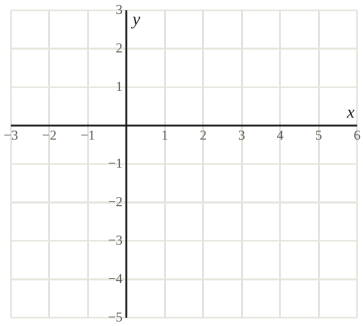


Standard form: Standard form: _____.

Intercepts: x-intercept: let _____ = 0. y-intercept: let _____ = 0.

Example 2

Find the intercepts of 3x − 4y = 12 and graph.



Part 2 · Partner practice

Partner practice

Find the intercepts of $-2x + y = 6$.